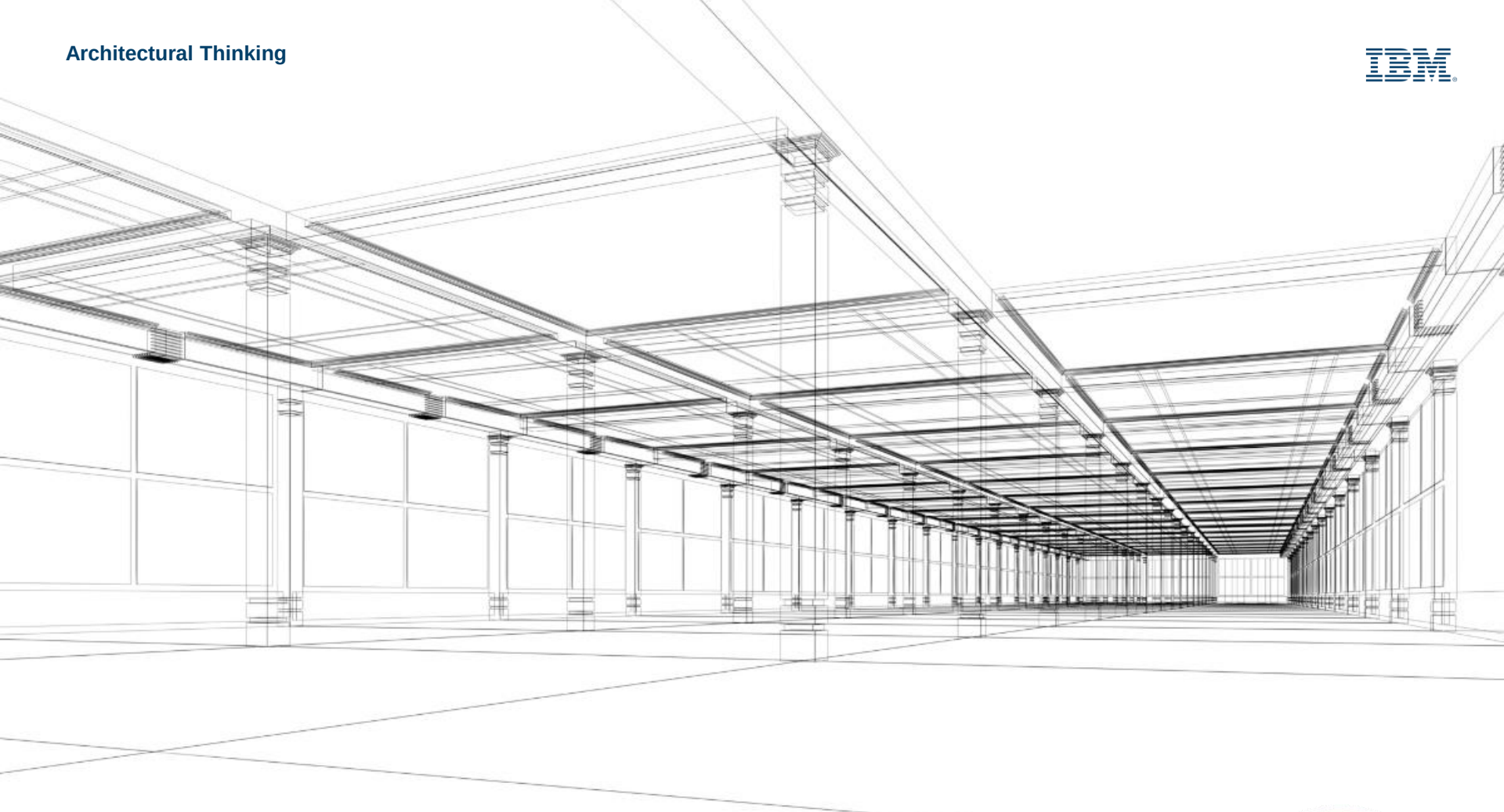




Operational Aspect





► **Introduction and objectives**

What is Operational Modeling (OM) and why do we need it?

How does OM evolve and how is it represented?

Constructing an Operational Model

Summary and references



At the end of this lecture, you should be able to:

- Understand and explain the:
 - Purpose of the operational aspect of architecture
 - Key concepts and artifacts of the operational aspect
 - Relationships between the functional aspect and the operational aspect
 - Techniques for creating Operational Models from organized sets of requirements
- Assist an experienced Architect in constructing and documenting the Operational Model
- Review an existing Operational Model document for completeness



Introduction and objectives

► **What is Operational Modeling (OM) and why do we need it?**

How does OM evolve and how is it represented?

Constructing an Operational Model

Summary and references

The Operational aspect is one of the two complementary aspects of architecture

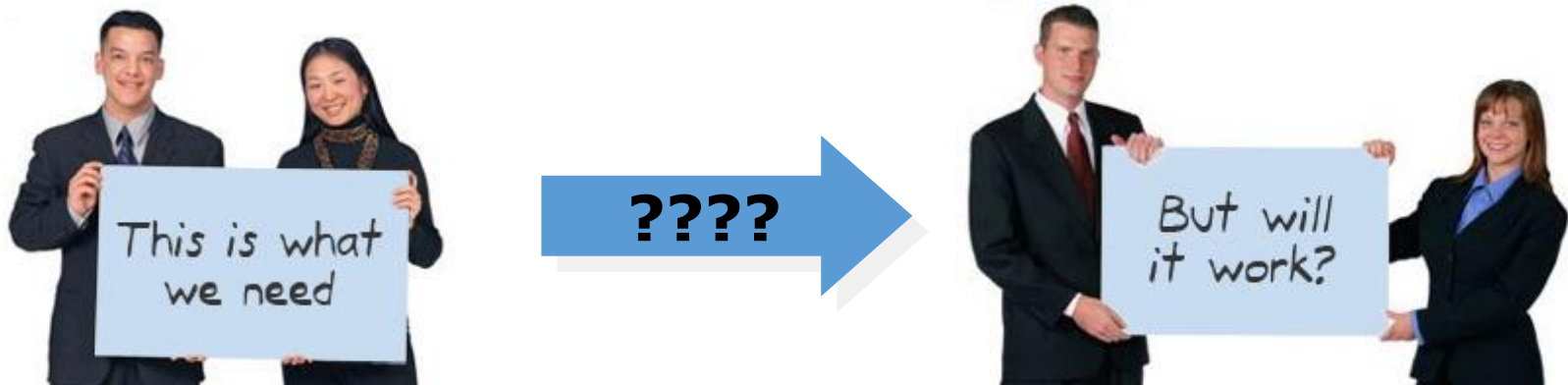


- The Functional aspect of Architecture <- **Previous Module**
 - Describe the structure of components (of all kinds) related to actual functionality,
 - ...their dynamic behavior (collaboration),
 - ...and their interfaces.
 - Embodied in the Component Model work product
- **The Operational aspect of Architecture <- This Module**
 - Describes what runs where (component placement).
 - ...and includes the network topology (hardware nodes, locations, etc.),
 - It ensures the achievement of the solution's service level characteristics (performance, availability),
 - ...and describes the management and operation of the IT system.
 - Embodied in the Operational Model work product



The design of a system's operational aspect may be challenging to develop or understand.

- Many competing concerns must be resolved:
 - NFR's: for example, performance, availability, manageability, security
 - Constraints: for example, affordability, standards compliance, existing Infrastructure
- Multiple activities must be coordinated:
 - **Detailed design:** capacity planning, software and data distribution, network, hardware, service management
 - **Commercial:** planning, pricing, procurement, service level agreement

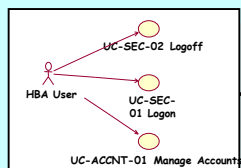


The design of a system's operational aspect may be challenging to develop or understand.

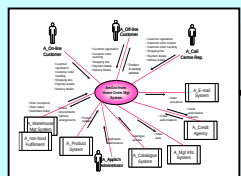
Operational Modeling helps ensure the system's non-functional (NFR) requirements are delivered, within all constraints



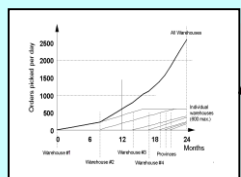
System Requirements



What do we have to do?
(Use Case Model)

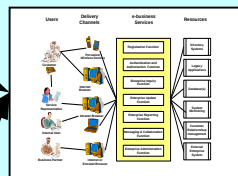


How do we fit in?
(System Context)

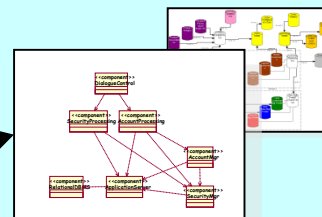


How good does it have to be?
(NFR Requirements)

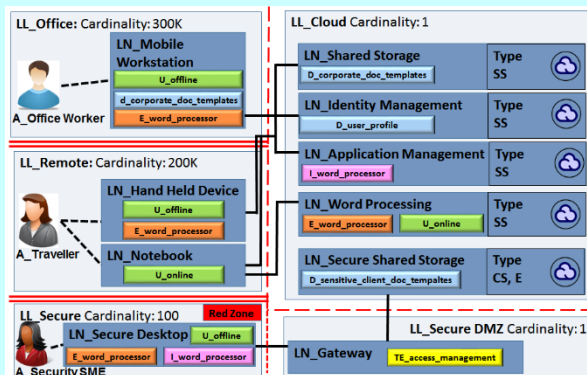
System Architecture



What is our approach?
(Architecture Overview)

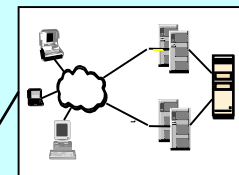


How is the application structured?
(Component Model & Data Model)



Where does everything go?
(Operational Model)

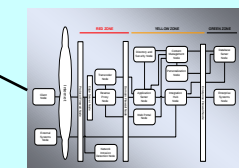
System Constraints



What is the current environment?
(Current IT)



What other constraints are there?
(Standards)



What experience do we have?
(Reference Models)

Designing for Performance and Capacity

Designing for Security

Assessing viability

Designing for Availability

Designing for System management



Time for a question



Select the correct statement from the following:

- A. The Operational Model focuses on the structure of the solution by identifying components and their collaborations.
- B. Service level requirements are delivered through the deployment of the components identified in the Operational Model.
- C. The Operational Model describes how components are supported to meet the service level requirements.
- D. The Operational Model is used as input when building the Location Model and the Node Model.

Module outline



Introduction and objectives

What is Operational Modeling (OM) and why do we need it?

► **How does OM evolve and how is it represented?**

Constructing an Operational Model

Summary and references

The Operational Modelling technique consists of multiple steps



1. Understand the **non-functional requirements**

- What NFRs need to be met ?

1M transactions per hour

2. **Conceptualize** from an **application** perspective (**ALOM**¹):

- **Identify** application level nodes and their placement.

Finance

Warehouse Management

Customer Relationship

3. **Conceptualize** the **Logical** Operational Model (**LOM**):

- **Determine** where all of the logical application components are deployed.
- **Determine** what logical **technical** components are required and where they will be deployed to support the logical application components.

ESB

Load balancer

Database

Application

Logical

4. **Transform** into the **Physical** Operational Model (**POM**)

- **Determine** real technologies to deliver all of these specifications.

SAP

Message Broker

F5 BIG-IP LTM-4200v

DB2

Physical

¹**Application Logical Operational Model (ALOM)** is **not** discussed in this lecture – see the Operational Modelling class for details on ALOM



- The Operational Model represents the system's “infrastructure architecture” using a variety of model elements, including:
 - The geographic structure of the **locations** and their **borders**, over which the IT system will be deployed and operated
 - The placement of the system's **nodes** into these locations
 - The deployment of the system's components across these nodes, using **deployment units**
 - The **connections** between nodes
 - The organization of the system's elements into **zones**
- In addition to this description of the deployed system, the OM also documents:
 - The overall **physical configuration** of the technologies and products necessary to deliver the functional and nonfunctional requirements of the IT system
 - Sizing and other **hardware specifications** for all the computers, storage devices, and network technologies

Operational Modeling (OM) notation used throughout this lecture and upcoming exercises.



Actors & Infrastructure

Type	Logical Example	Physical Example
<u>L</u> ocation	LL_Office	PL_IBM Bolumbia Head Office
<u>N</u> ode	LN_Mobile Workstation	PN_Apple Laptop
Human <u>A</u> ctor	A_Office Worker  	
System <u>A</u> ctor	A_Respondent PC 	

Connections

No permitted connection

External intermittent connection

Internal high speed connection

External high speed connection

Cloud

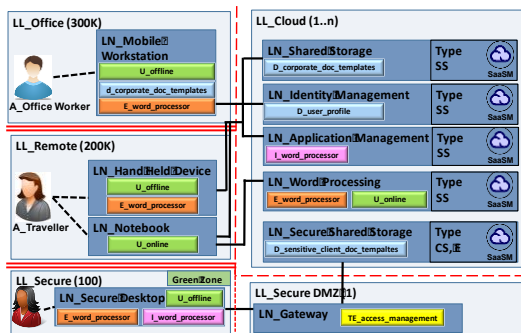
Type	Example
Offering	 On-premises  IaaS (Infrastructure)
	 SaaSM (Software) Managed  SaaSH Hosted
	 External/Web Resource
Support	SS - Standard Service Level Agreement(s) CS - Custom Service Level Agreement(s)
Data	E - Encrypted, NE - Non-Encrypted, AU - Audited, AR - Archiving Required

Deployment Units

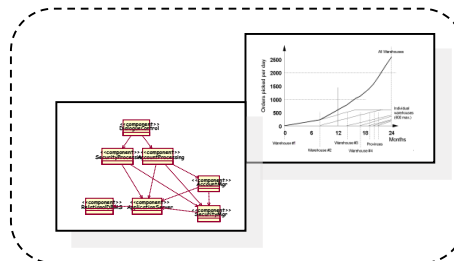
Capital letter = Master copy
and lower case = copy

Type	Short ID	Long ID
Presentation	U_<X>	U_Name
Execution	E_<X>	E_Name
Installation	I_<X>	I_Name
Data	D/d_<X>	D_Name
Technical	TE_<X>	TE_Name

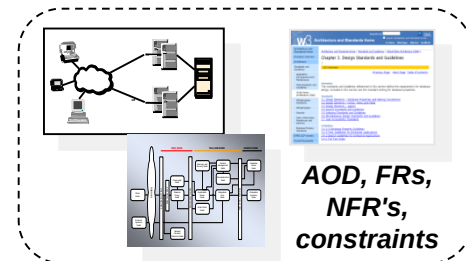
Operational Model inter-dependencies with other artefacts



**Where does everything go?
(Operational Model)**

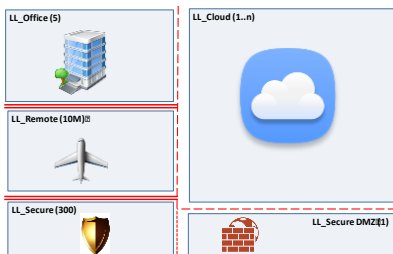


Component Model



**AOD, FRs,
NFR's,
constraints**

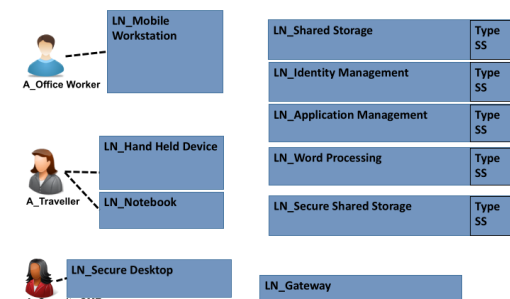
Based on the CM



**Where can we put things?
(Location Model)**

PresentationDU		IT Skill level	Location	Frequency	Type
U_Inventory	Inventory Check	Can be trained	Mobile	N/A	N/A
U_SMS	Product Catalog	N/A	N/A	Weekly	Batch
2					
Data DU	Scope	Volatility	Size	Lifetime	Master or Copy
D_Inventory	Store	High	Small	<= 1 day	N/A
D_Cat	Entire product range	Low	Large	Long lived	M
Execution DU		Frequency	Processing Load		
E_Submit/Inv_Update		Daily (once for every store)	Light		
E_Consolidate		Daily	Heavy		

**What do we need to deploy?
(Deployment Unit Model)**



**Where should we
deploy things?
(Node Model)**

**These contained artifacts give
us a way of "building up" the
Operational Model...**



LL_Office (5)



LL_Remote (10M)



LL_Secure (300)



LL_Cloud (1..n)



LL_Secure DMZ (1)





Presentation

Entry point used by actors to access a components functionality e.g. live web interface.

Data

The data used by components e.g. user profile.

Execution

The runtime execution aspect of a component e.g. 3 instances of a word processor running in 3 different user accounts on the same mobile workstation.

Installation

Files required to install the component e.g. executable, configuration and data.



List deployment units for each component

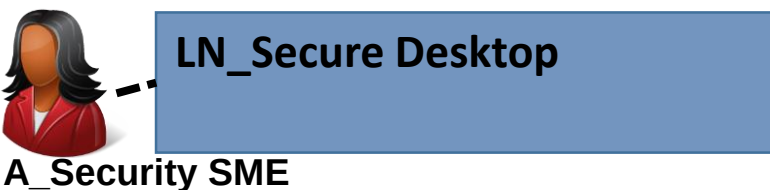
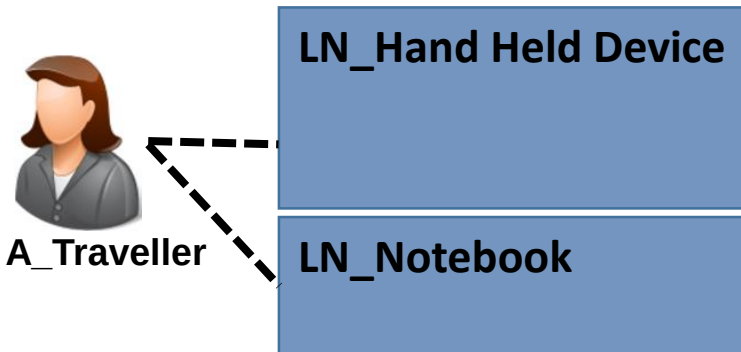
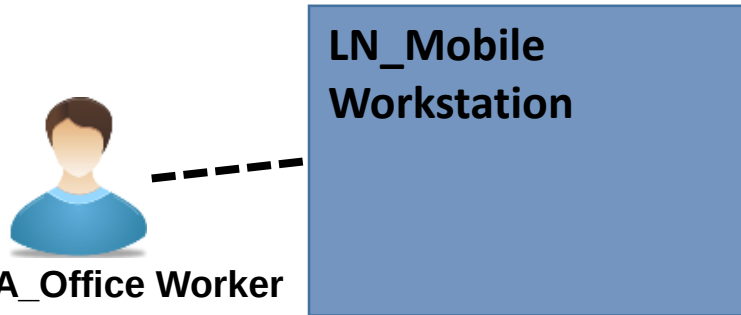
Component (from the CM)	Presentation Deployment Unit	Data Deployment Unit	Execution Deployment Unit
Customer Relationship Management	U_CRM	D_CRM_Data	E_Update_CRM_Data E_Consult_CRM_Data
Stock Management	U_SMS U_Inventory	D_Inventory_Upd	E_Submit_Inv_Update E_Consolidate_Inv_Update

DUs need to be described regarding their non-functional requirements which determine their placement and grouping.

PresentationDU	Actor	IT Skill level	Location	Frequency	Type
U_Inventory	A_Inventory_Check	Can be trained	Mobile	N/A	N/A
U_SMS	A_SMS	N/A	N/A	Weekly	Batch

Data DU	Scope	Volatility	Size	Lifetime	Master or Copy
D_inventory	Store	High	Small	<= 1 day	N/A
D_Cat	Entire product range	Low	Large	Long lived	M

Execution DU	Frequency	Processing Load
E_Submit_Inv_Update	Daily (once for every store)	Light
E_Consolidate	Daily	Heavy



LN_Shared Storage	Type SS
LN_Identity Management	Type SS
LN_Application Management	Type SS
LN_Word Processing	Type SS
LN_Secure Shared Storage	Type SS



We decide where the Deployment Units WILL go Following One of Two Paths



PresentationDU	Actor	IT Skill level	Location	Frequency	Type
U_Inventory	A_Inventory_Check	Can be trained	Mobile	N/A	N/A
U_SMS	A_SMS	N/A	N/A	Weekly	Batch

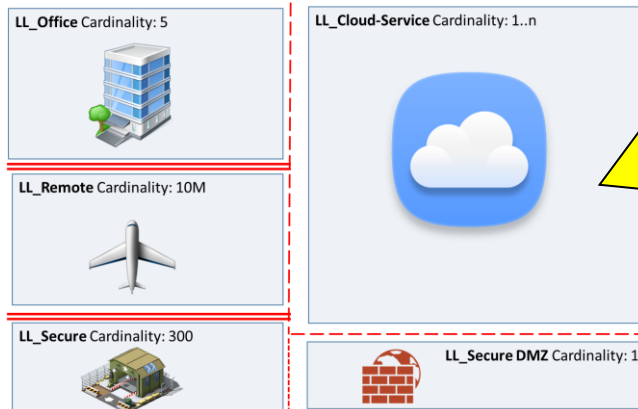
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D_Cat	Entire product range	Low	Large	Long lived	M

Execution DU	Frequency	Processing Load
E_Submit_Inv_Update	Daily (once for every store)	Light
E_Consolidate	Daily	Heavy

What do we need to deploy?

(Deployment Unit Model)

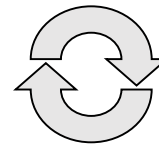
Path 2 forward engineered where DU's are placed into locations where they are accessed by Actors



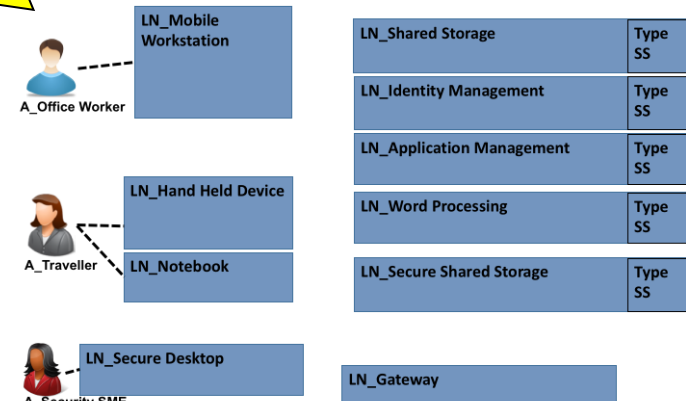
Where can we put things?
(Location Model)

Path 1 using an asset where DU's are associated with a predefined node model and nodes are deployed into their locations. We will use this approach for the worked example.

1.1 Deploy DUs



Iterate!

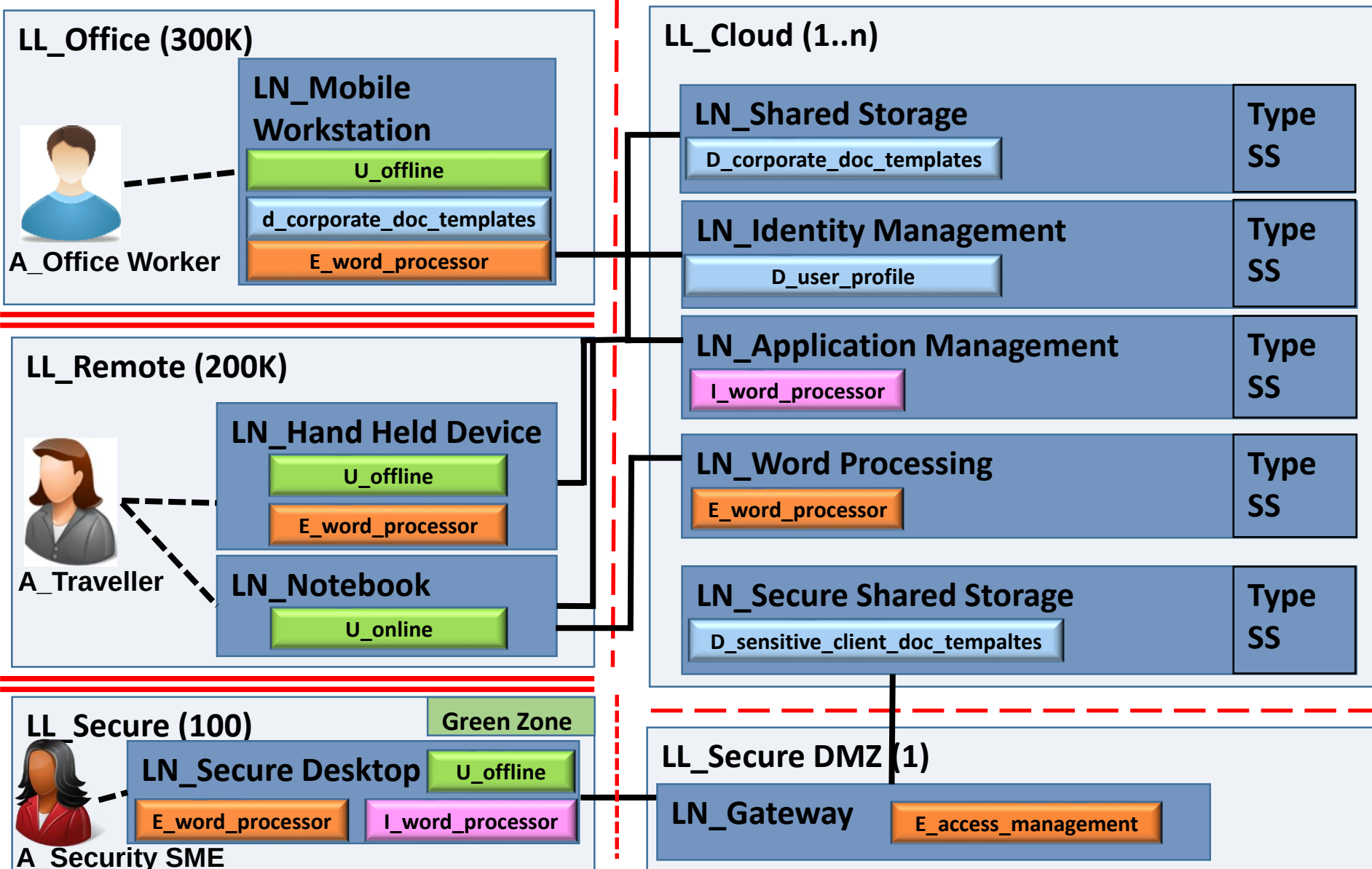


1.2 Deploy Nodes

2.2 Deploy DUs

Path 2 nodes are formed at locations that contain DU's

Logical Operational Model - shows the static relationships between nodes, DUs, connections and interactions on a geographic background





1. Understand the **non-functional requirements**

- What NFRs need to be met ?

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1M transactions per hour

Finance

Warehouse Management

Customer Relationship

ESB

Load balancer

Database

Application

Logical

SAP

Message Broker

F5 BIG-IP LTM-4200v

DB2

Physical

The LOM documents the placement and interaction of the application's DUs onto nodes distributed across the system's locations



The standard technique involves:

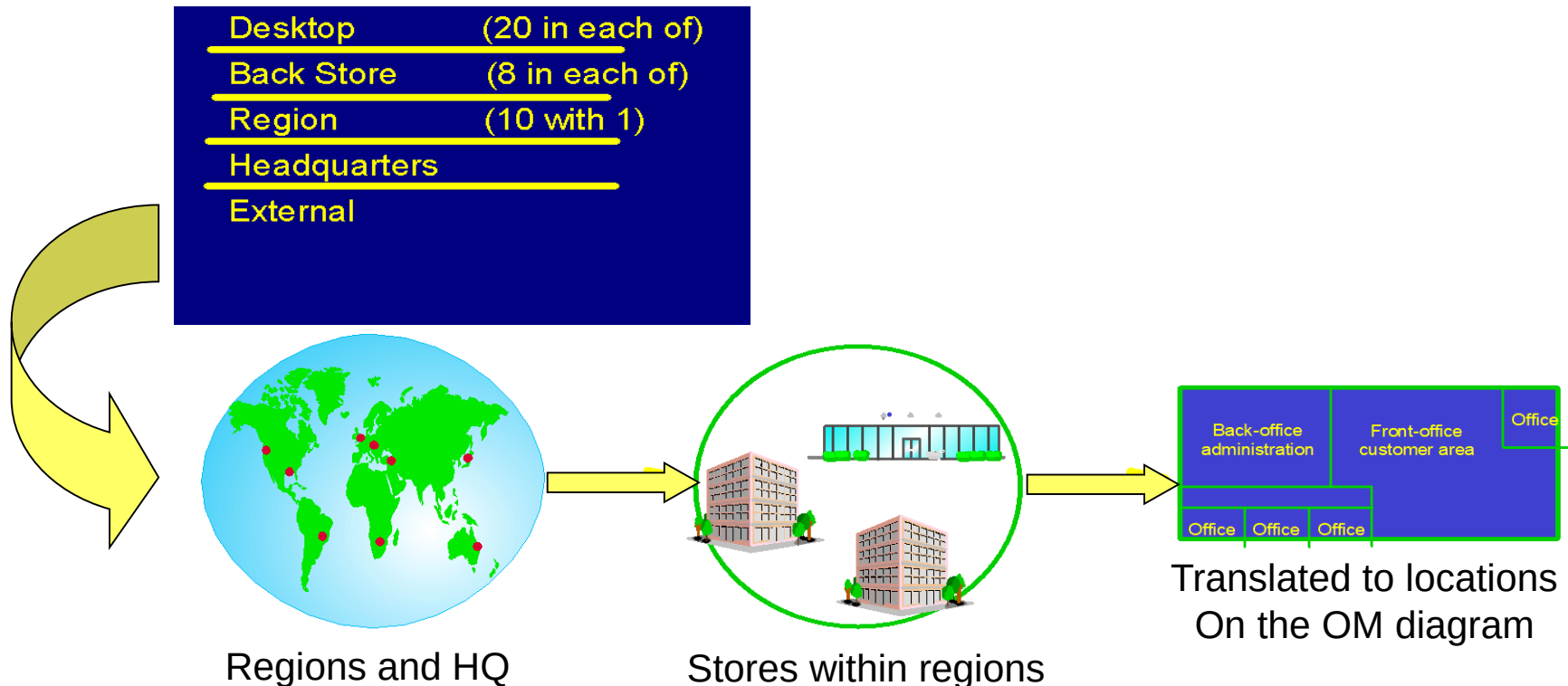
- Identifying potential locations and nodes where deployment units might end up being placed, and establishing where the system's actors will be located
- Placing deployment units onto application logical nodes in locations, based on a variety of considerations such as service management
 - Presentation DUs
 - Data DUs
 - Execution DUs
 - Installation DUs
- We derive connections from the relationships (interactions) between deployment units
- (Confirming the design using architecturally significant walk-throughs)

Step 1: The overall structure of the LOM is largely based on the locations over which the solution will be deployed.



The location is based on the information found in:

- Business Roles & Locations
- System Context
- Current IT Environment



Step 2: Then identify Candidate Logical Nodes



Inputs

System Context

Use Cases

NFRs

Component Model

Key Questions to ask for node discovery

- Is there different functionality or placement across locations?
- Are applications accessed by actors in different locations?
- Are there groups of applications with different NFRs/SLAs?

These are NOT logical nodes

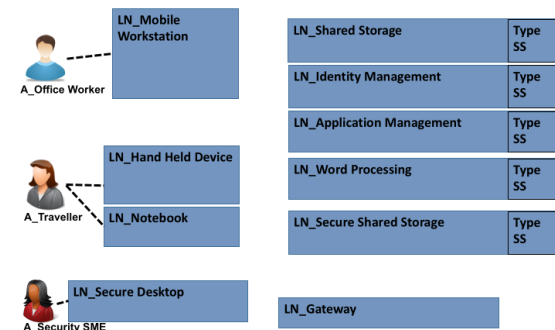
Hardware devices e.g. routers

Named hardware products

LANs and WANs



Output



Step 3: We have to understand what it is we need to deploy, via the DEPLOYMENT UNIT Model

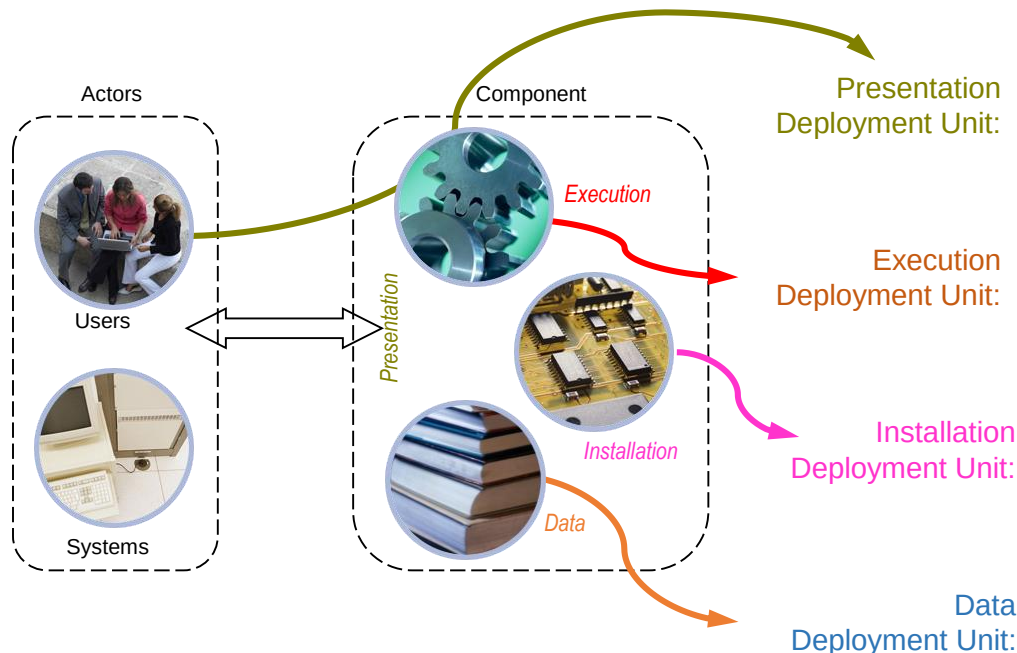


- DUs give us a mechanism for:
 - Understanding the nonfunctional requirements placed on the system's components
 - Deciding where best to place the various aspects of the system's components
 - Tracing the system's requirements to its design

The nonfunctional requirements of the component's interactions are described in terms of DU – DU relationships.

- Typical relationships analyzed include:

- Presentation ↔ Execution
- Execution ↔ Execution
- Execution ↔ Data
- Presentation ↔ Data



Nature of Access Point	Actor using Access Point	Location	Required Hours	Concurrent Active
------------------------	--------------------------	----------	----------------	-------------------

Txn Type	Usage 'Window'	Number per user	95% response time
----------	----------------	-----------------	-------------------

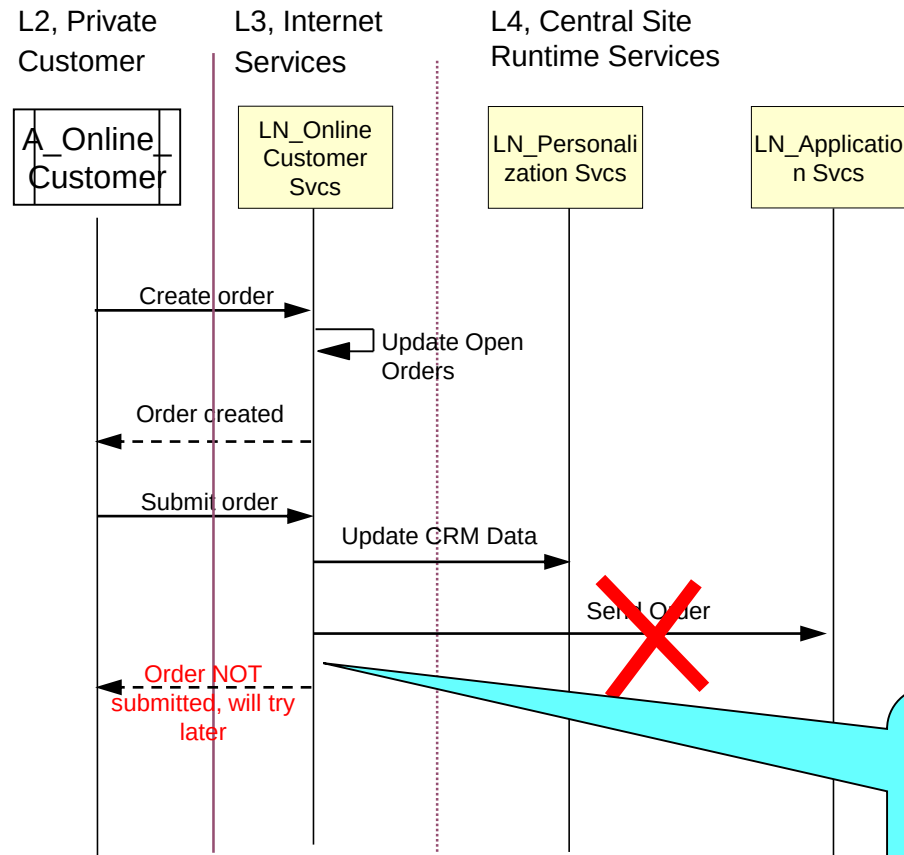
Size	Release Frequency
------	-------------------

Data Type	Unit/ record Size	Number records	Volatility	Currency
-----------	-------------------	----------------	------------	----------

Step 4: Operational Modeling helps ensure the system's non-functional requirements are delivered, within all constraints...



A dynamic walkthrough technique can be used to demonstrate that various Use Cases are indeed supported by the OM and often able to trigger improvements to business processes.



Highlights the need for LN_Online Customer Services to hold a temporary master of the submitted order – NOT shown in the OM relationship diagram!



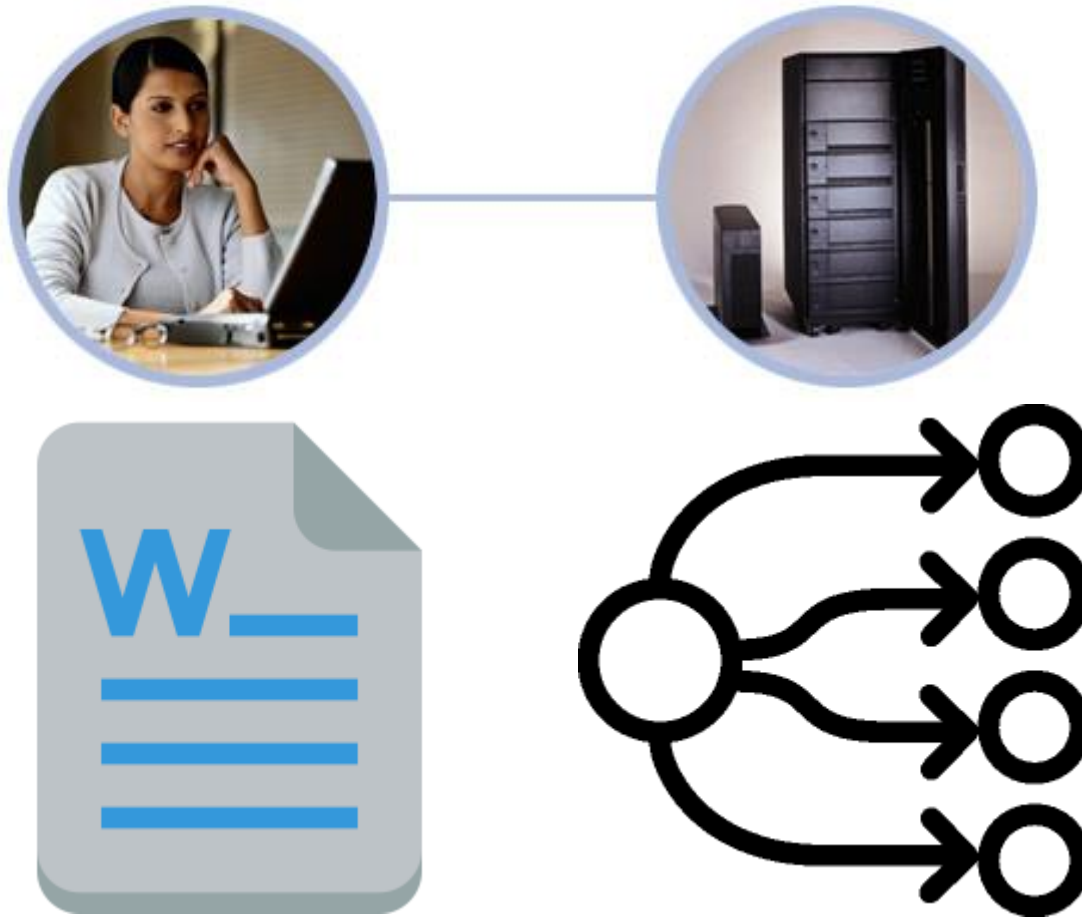
Time for a question



The typical views of an Operational Model include the Physical Operational Model, and the:

- A. Logical Operational Model
- B. Component Operational Model
- C. Technical Operation Model
- D. Deployment Operation Model

Example: Where and how do we “deploy” a Word Processor ?





What are the use cases that need to be met ?

Story 1: Sales Manager working from a IBM office creates a statement of work using an IBM approved template.



Story 2: Project Manager while traveling creates a Project Change Request using a IBM approved template.



Story 3: Architect working from a IBM office in the public sector creates a confidential design using a template from an IBM asset that is not allowed to be stored outside of local country



**Simple
Right ?
Lets see...**

First we model the landscape from a logical perspective and then from a physical perspective



Logical Examples

Functional Component
Word Processor



Technical Component
Identity Management



Location
Employee Office



Deployment Unit
Data: SOW Template



Node
Mobile Workstation



Physical Examples

MS Word 2016

MS Active Directory

IBM Head Office

**IBM Master SOW
Template.docx**

Macbook

Word Processor Example - Logical Location Model



LL_Office (5)



LL_Remote (10M)



LL_Secure (300)



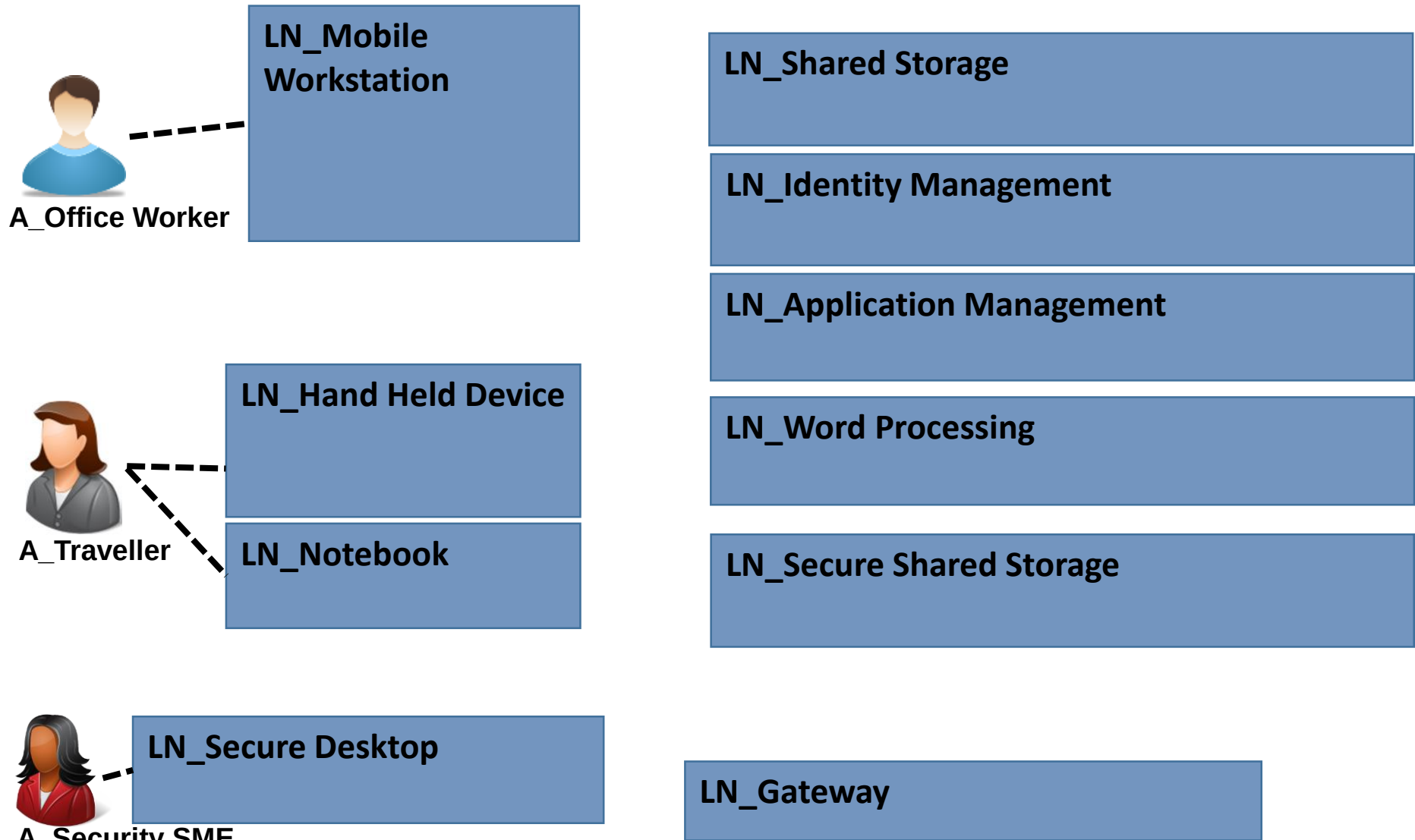
LL_Cloud (1..n)



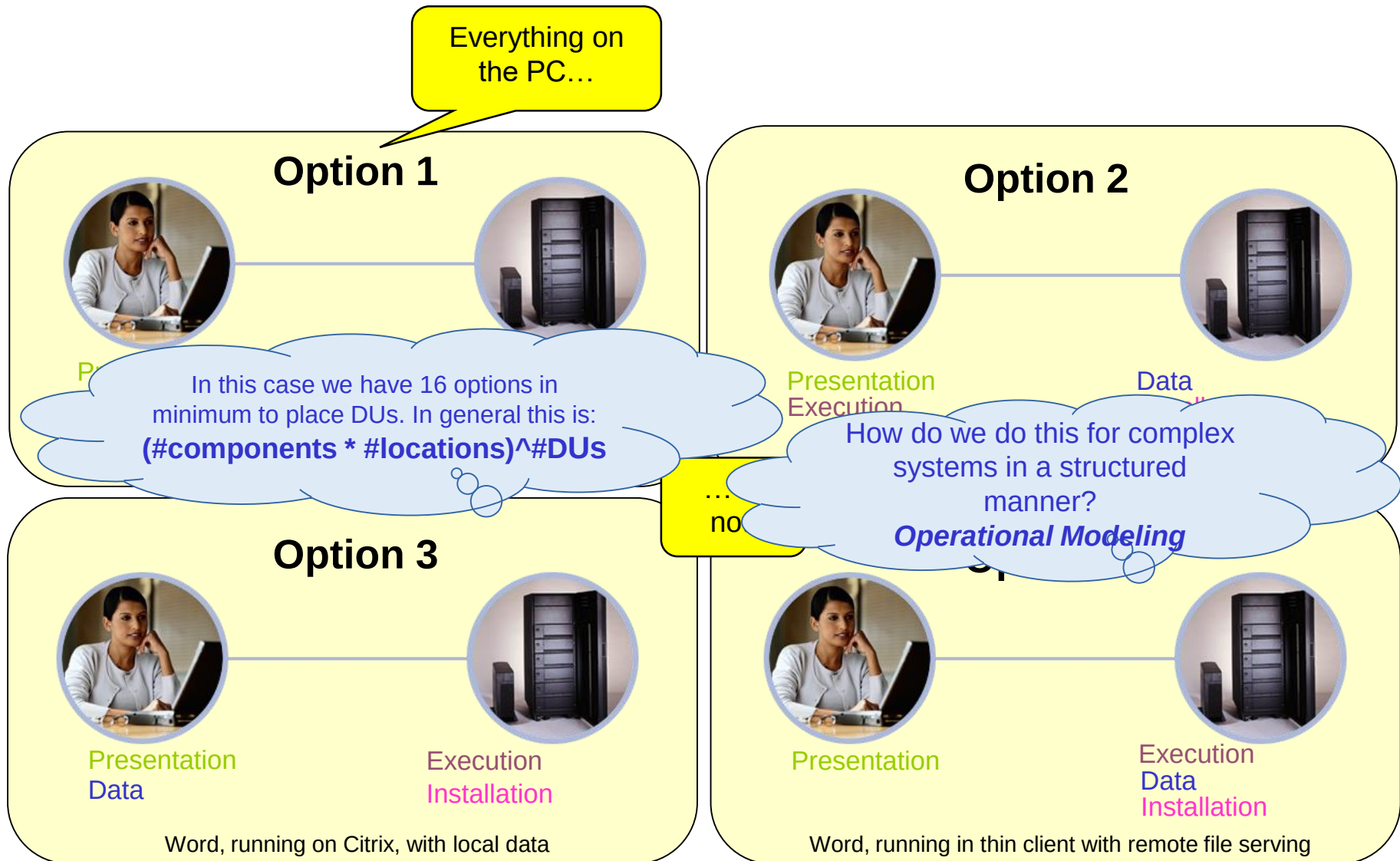
LL_Secure DMZ (1)



Word Processor Example - Logical Node Model



There are many ways of deploying a single component into a simple system



Based on our use cases we get these deployment units?



Word Processor

Deployment Units

Presentation

U_offline

U_online

Installation

I_word_processor

Data

D_corporate_doc_templates

D_sensitive_client_doc_templates

D_user_profile

Execution

E_word_processor

E_access_management

More
things to
consider

Worked Example – Word Processor Logical Operational Model Reference Architecture Asset Example showing existing nodes



LL_Office (300K)



A_Office Worker

LN_Mobile Workstation

LL_Remote (200K)



A_Traveller

LN_Hand Held Device

LN_Notebook

LL_Secure (100)

Green Zone



A_Security SME

LN_Secure Desktop

LL_Cloud (1..n)

LN_Shared Storage

LN_Identity Management

LN_Application Management

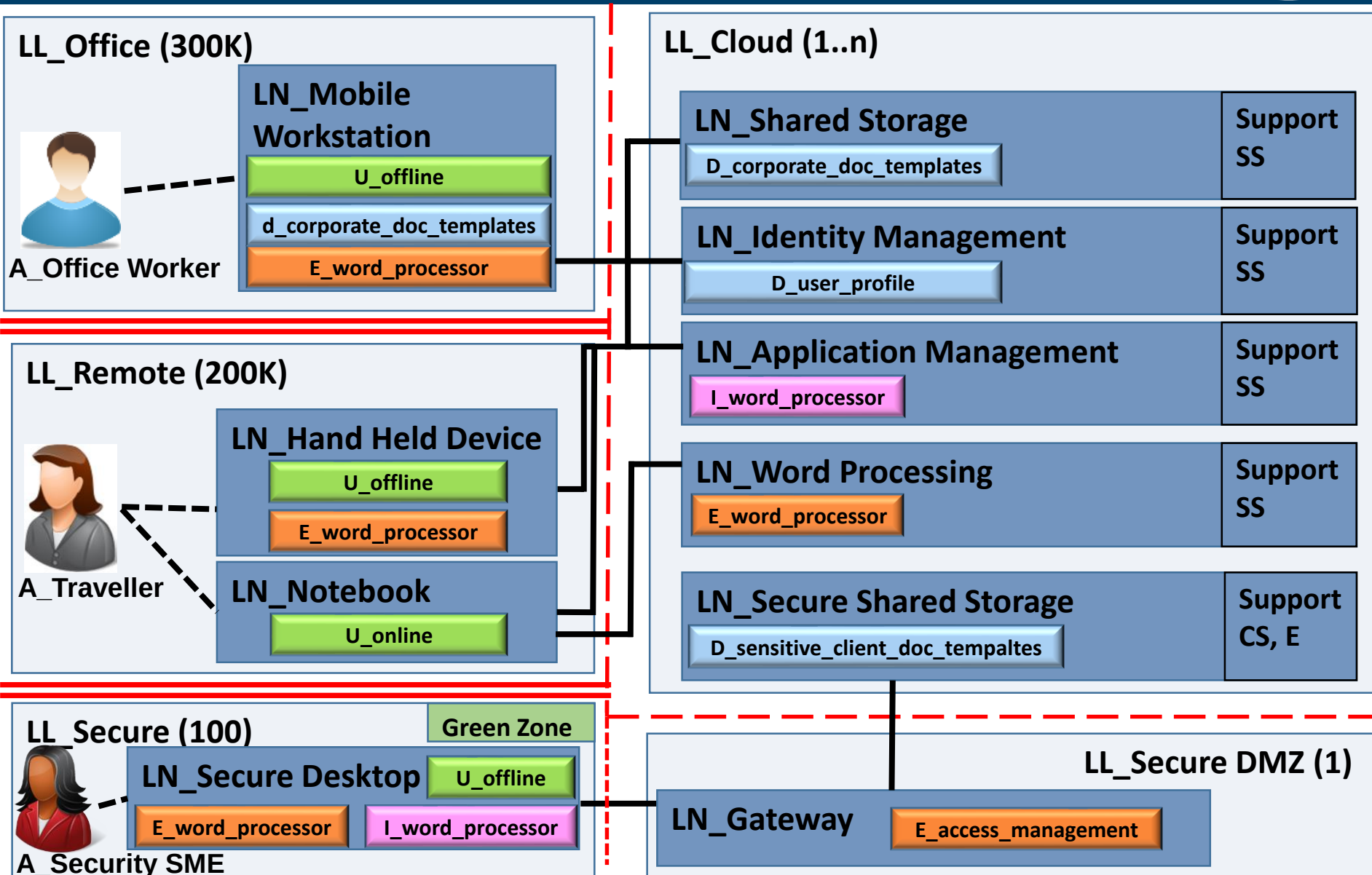
LN_Word Processing

LN_Secure Shared Storage

LL_Secure DMZ (1)

LN_Gateway

Worked Example – Word Processor Logical Operational Model Place Deployment Units onto Nodes Following Asset





BREAK



1. Understand the **non-functional requirements**

- What NFRs need to be met ?

2. **Conceptualize** from an **application** perspective (**ALOM**¹):

- **Identify** application level nodes and their placement.

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Warehouse Management

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F5 BIG-IP LTM-4200v

DB2

Physical

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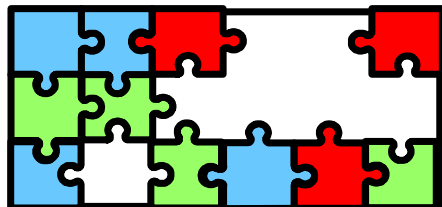
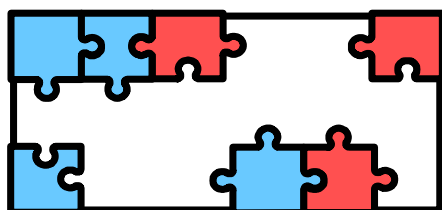
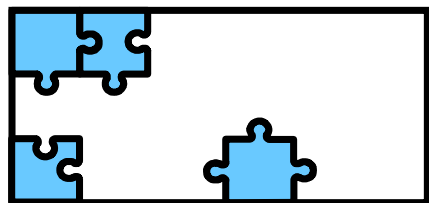
The POM transforms the logical system architecture into a set of product and technology specifications



The standard technique involves four steps:

- Identifying your role in the product selection process and the definition of the selection criteria
- Performing the necessary product selections for the nodes and connections
- Defining the solution details and describing how technologies will be put together to deliver the specific SLAs
- Rationalizing the solution architecture to get to a sized POM

Product selection is the most important part when we build the physical operational model.



- ✓ **1. Identify the “Givens”**
 - In a client engagement
 - Usually defined in the client’s IT standards, existing Technical Architectures or current IT environment. In all cases, from the beginning you will have ensured the LOM is capable of being implemented in these technologies
 - In an IBM proposal
 - IBM product, *if* they meet the requirements!
- ✓ **2. Spot the “Obvious” products**
 - “Easy to choose”, such as:
 - “Only one product/vendor meets the specs.”
 - “We all *know* this will be product/vendor ABC”
 - “We *could* do a full vendor evaluation study, but it is more efficient to choose this product/vendor now”
 -
 -
- ✓ **3. Look at the “Adjacent” components**
 - The ones that are the most difficult to choose



Physical Nodes

- Hardware technologies and products
 - Overall hardware configuration
 - Hardware specifications, such as processor speed, memory, disk configuration, etc.
 - for example, SystemP, SystemX...
- Software technologies and products
 - Software product specifications for operating system, database, middleware, directory services, security, etc., including versions
 - for example AIX Version X.x, WebSphere Application Server Liberty Profile, etc...
 - Detailed configuration of software, such as the need for multiple instances of a software product on a computer

Physical Connections

- Technology specific, including
 - Physical networks, protocols, network bandwidth, latency, etc.
 - for example, 100BaseT, Fiber Optic, T1, T3, 1/10/100 Gbps Ethernet, TCP/IP, etc.



Cloud Services

- Cloud deployment model
 - **Public**: Shared infrastructure that lowers the cost per unit for all customers.
 - **Private**: Provides dedicated infrastructure configured for flexibility and scalability.
 - **Managed**: Takes care of infrastructure management and maintenance tasks.
 - **Hosted**: Provides flexibility to self source infrastructure management and maintenance support.
 - **Hybrid**: Mix of using shared infrastructure and dedicated infrastructure depending on NFR's.
- Cloud services model
 - **IaaS** : Infrastructure as a Service - provides raw infrastructure resources in a virtualised environment e.g. IBM Bluemix.
 - **PaaS**: Platform as a Service - provides an container/runtime environment with supporting services to build solutions e.g. IBM Bluemix.
 - **SaaS**: Software as a Service – provides application functionality as a service e.g. IBM Verse.
 - **BPaaS**: Business Process as a Service – provides a means to increase organization efficacy by outsourcing business processes to a automated runtime e.g. through the use of IBM Blueworks Live



Traditional LOM to POM transformation

A distinction can be drawn between Physical Level Operational Models that emphasize



A distinction can be drawn between Physical Level Operational Models that emphasize:

- Component Selection (Unsize POM): Identifying the technologies and products, and the overall (qualitative) way in which they will be put together:
 - Identify the “givens”
 - Spot the “obvious” products
 - Look at the “adjacent” components
- Configuration (Sized POM): Defining the (quantitative) details of the solution, including the way these technologies will be put together in order to deliver the specific SLRs of the OM’s specification:
 - Performance and capacity
 - Availability
 - Security
 - Service management

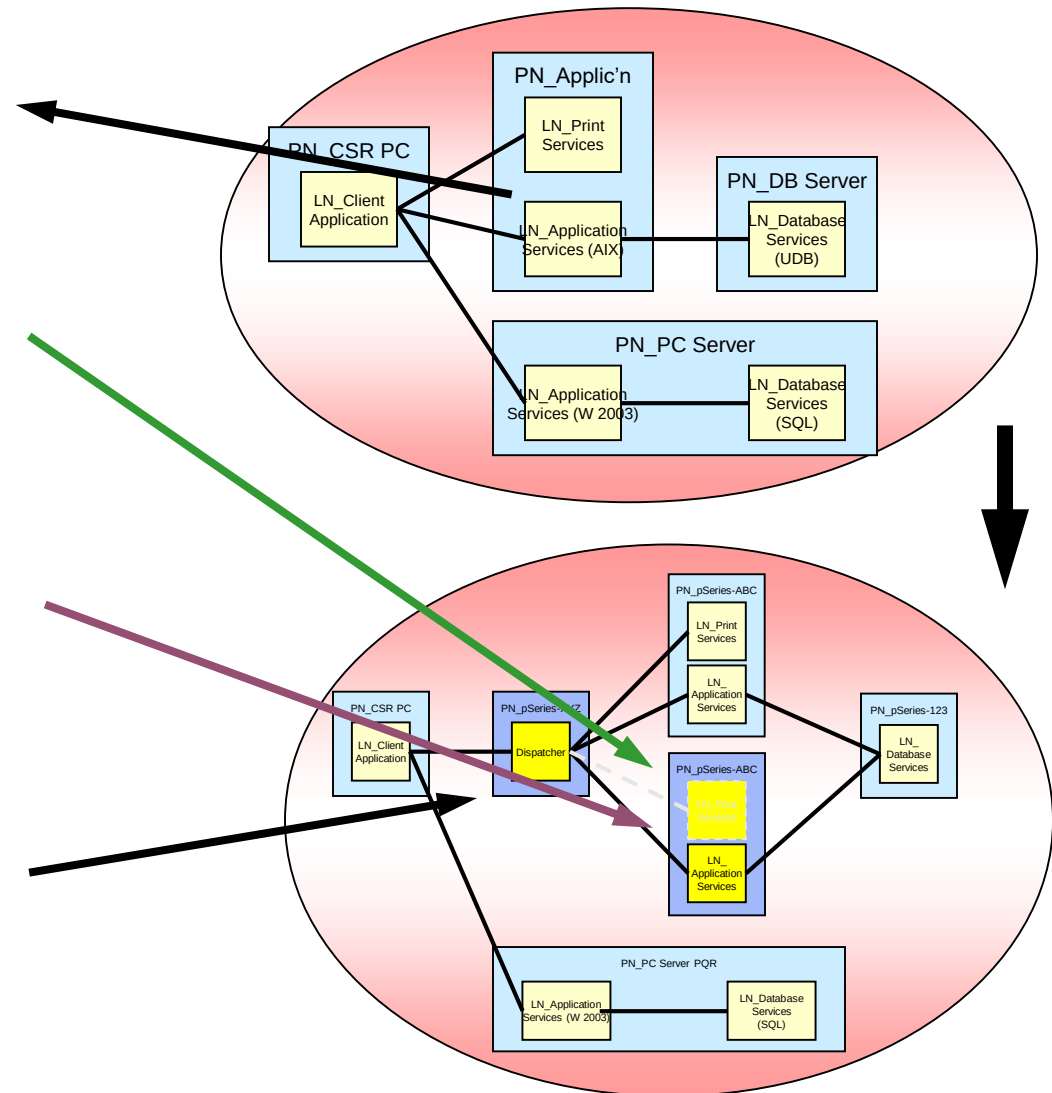
Rationalizing the design involves a process of SIZING the POM, so that performance & other characteristics can be achieved.



For example, the “PN_Applic’n” physical node will deliver:

- The availability requirements of the Print Services, by being implemented as two physical pSeries machines, with a failover capability between the “live” and “standby” machines
- The throughput requirements of Application Services, by being implemented as TWO physical pSeries machines, together with a “dispatcher” physical node.

This, of course, has an influence back on the POM, through the introduction of a “LN_Dispatcher” logical node





Cloud LOM to POM transformation



1. **Plan** - Review the Cloud reference architectures and determine if one is a good match for the NFR requirements or tailor as required.
2. **Define** - the type of cloud deployment model (CDM) i.e. public, private, hybrid, that meets the solution NFR's and constraints.
 - Logical Service Models e.g. IaaS, PaaS, SaaS, BPaaS
 - Logical Services e.g. Middleware, Database, Storage
3. **Select** – a cloud provider that meets NFR and CDM requirements with a sizing/pricing model that provides the best match for the solution.
4. **Configure** – Using NFR's, LOM and cloud reference architecture as inputs
 - Configure the required physical cloud services offered within the chosen providers CDM.
 - Configure performance, capacity, and availability by configuring physical cloud services sizing/service level requirements (SLR's)
 - Understand licensing model including enterprise level agreements for software/services installed within IaaS and PaaS.

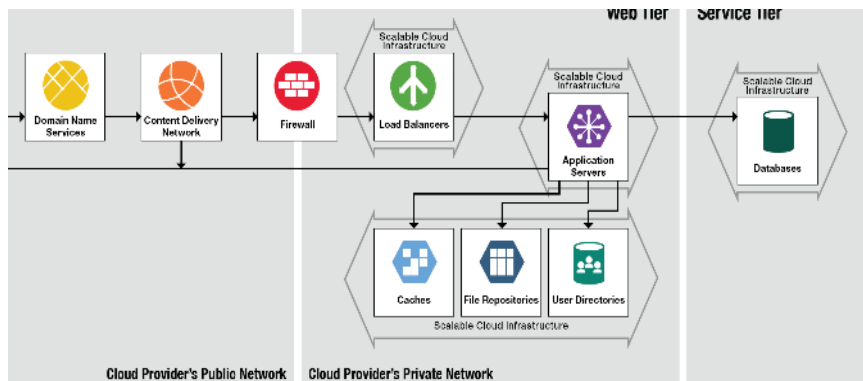




Selecting a cloud reference architecture helps leverage lessons learnt and best practices but you may still need to tailor to your needs or in the case none fit the solution define your own

Cloud Council Reference Architectures

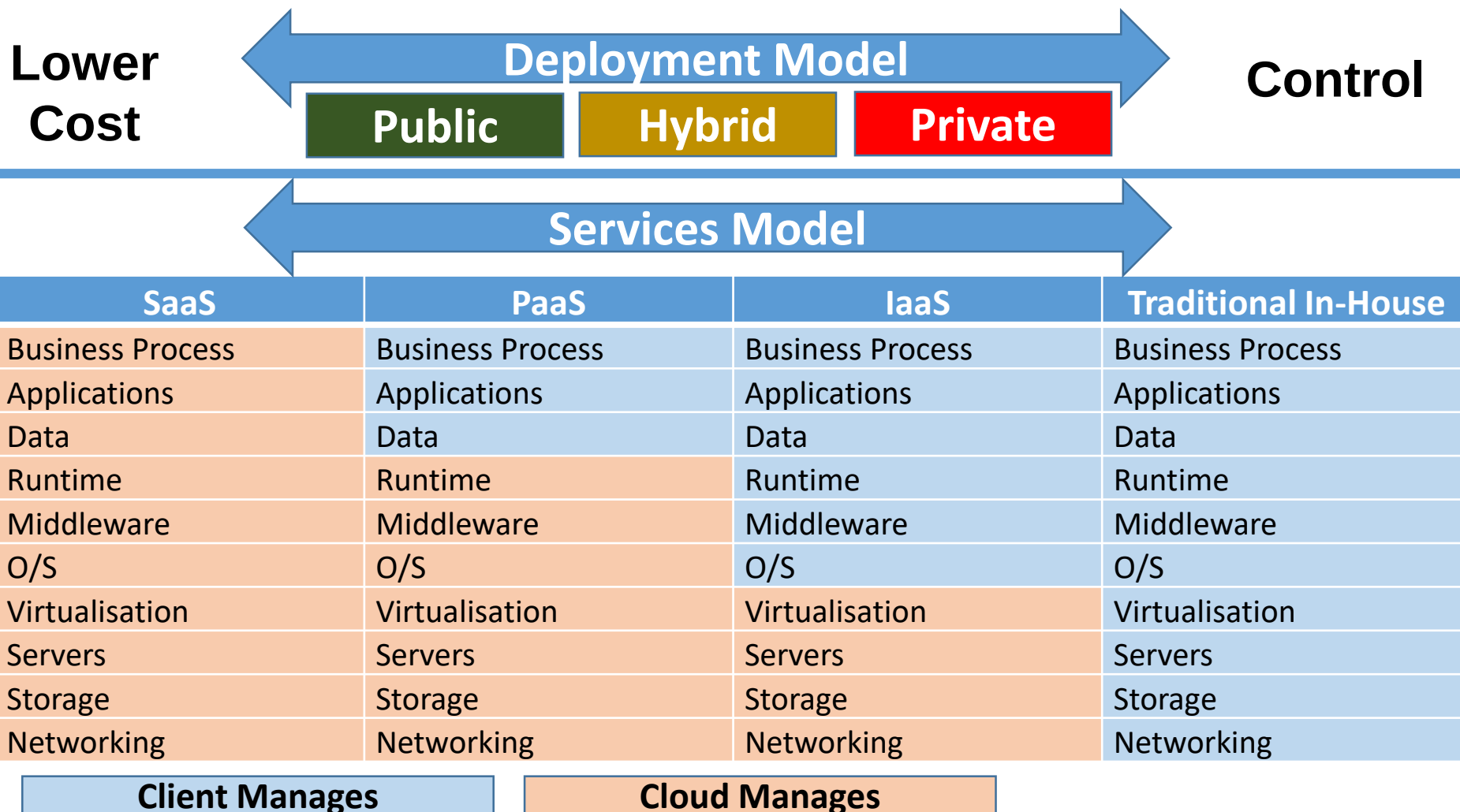
- [Cloud Customer Architecture for Securing Workloads on Cloud Services](#)
- [Cloud Customer Architecture for API Management](#)
- [Cloud Customer Architecture for Hybrid Integration](#)
- [Cloud Customer Architecture for Enterprise Social Collaboration](#)
- [Cloud Customer Architecture for e-Commerce](#)
- [Cloud Customer Architecture for IoT](#)
- [Cloud Customer Architecture for Big Data and Analytics](#)
- [Cloud Customer Architecture for Mobile](#)
- [Cloud Customer Architecture for Web Application Hosting](#)



Cloud@POM – Define the cloud deployment model that meets the solution requirements and constraints



Architectural decisions need to be made to select the required deployment and service model(s) based on the NFR's and constraints while balancing value for the Customer.





- Some considerations include:
 - Availability of the cloud platform within the client's jurisdiction.
 - Enterprise strength of the services offered by cloud providers, in particular security, availability, backup and disaster recovery.
 - The level of support offered by the cloud provider
 - Network connectivity to the cloud provider's data centre. One of the disadvantages of a cloud based solution is the potential for increased latency.
 - Content of the cloud provider's service catalogue (e.g. operating system, databases, middleware, ect).



Many others!





1. Need to select a physical service for each logical services identified in the LOM i.e. Database -> DashDB
2. Define the size of the cloud services and as you can see the major cloud providers use a T-shirt sizing model and you need to consider the NFR's to select the “T-shirt” the best fits the solution while keeping customer value in mind

Microsoft Example

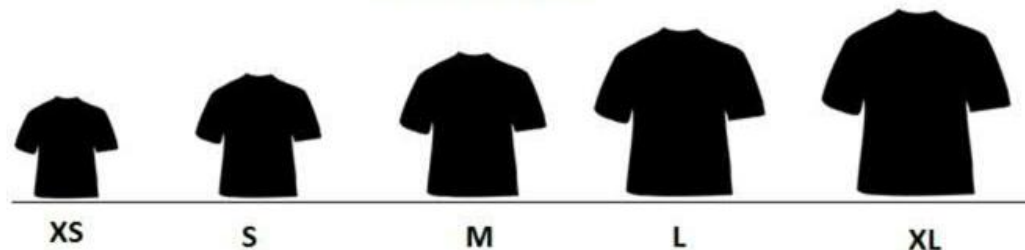
SKU Family	ACU/Core
ExtraSmall	50
Small-ExtraLarge	100
A5-7	100
Standard_A1-8v2	100
Standard_A2m-8mv2	100
A8-A11	225*
D1-14	160

IBM Example

T-Shirt	vCPU	RAM (GB)	HD (GB)	Price/Hr
S	1	2	12	\$0.21
M	2	4	25	\$0.42
L	4	8	50	\$0.84
XL	8	16	100	\$1.68
XXL	16	32	200	\$3.36

SAP Example

SAP HANA Sizing
Fit-For-Purpose



The review process leverages transaction walkthroughs to test the solution's robustness



At every level of the OM, “transaction walkthroughs” using the OM walkthrough diagram are a powerful means of verifying whether the OM will work

- Analyze each transaction from start to finish following the flow of information through each physical node and connection
- Visualize what would happen if there was a failure on some part of the system:
 - Would the failure be detected?
 - Would the end user be notified of the failure?
 - How will the transactions and data be recovered?
 - Who will be responsible for repair of the component?
- Analyze how the failure would impact Service Level Agreements

Module outline



Introduction and objectives

What is Operational Modeling (OM) and why do we need it?

How does OM evolve and how is it represented?

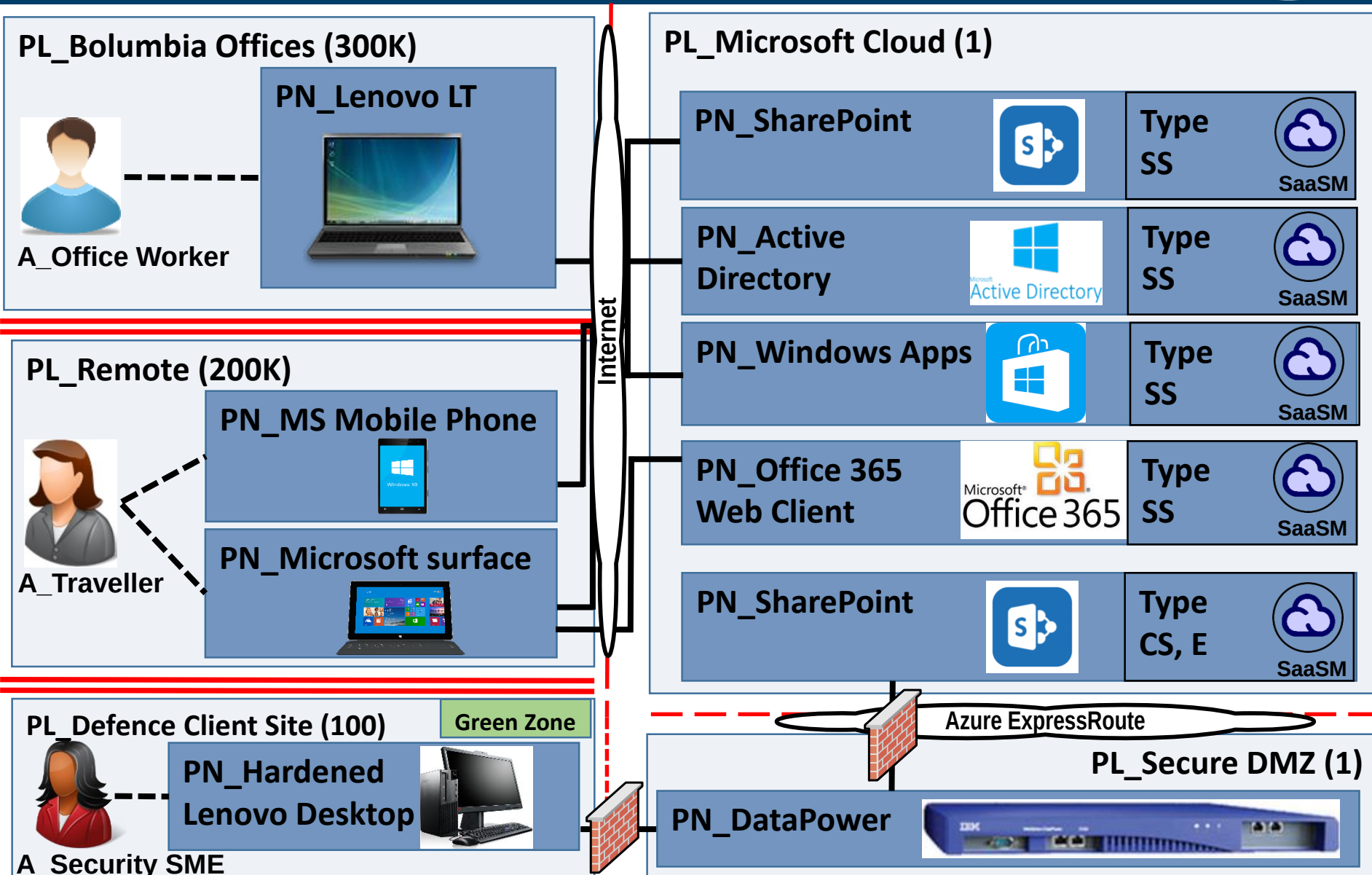
► **Constructing an Operational Model**

Summary and references



Lets transform our LOM into a POM

Worked Example – Word Processor Physical Operational Model



Module outline



Introduction and objectives

What is Operational Modeling (OM) and why do we need it?

How does OM evolve and how is it represented?

Constructing an Operational Model

► **Summary and references**



The operational aspect of architecture:

- Focuses on the placement of the solution's DUs across the organization to achieve the business requirements of performance, availability, and other nonfunctional requirements
- Is addressed through the creation of the Operational Model at 2 different viewpoints:
 - The Logical Operational Model, which focuses on the distribution style of the application and the technical components required to support the application components
 - The Physical Operational Model, which details which combination of products, hardware and or cloud based services would deliver on the specification defined through the LOM



Enterprise Solutions Structure – ESS

- http://ieeexplore.ieee.org/xpl/freeabs_all.jsp?arnumber=5387080

Cloud Computing

- <https://www.ibm.com/developerworks/learn/cloud/index.html>

Cloud council

- <http://www.cloud-council.org/>



- A number of supporting documents can be found on MethodWeb including:
 - Guidance / Technique Papers
 - Developing the logical application operational model
 - Developing the logical operational model
 - Developing the physical operational model
 - Designing for Availability
 - Designing for Performance and Capacity
 - Designing for Security
 - Designing for Systems Management
 - Architecture Domain and related Work Product (WP) Descriptions
 - Other background papers on w3 from various sources

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